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MLA0401- Deeplearning.

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Question- 1

A coin tossed 10 times and the outcomes are: HHTHHHTHHT. find the probability of geeting Heads' (p) using Maximum Likelihood Estimation (MLE)

solution:-

Given

- Total number of tosses (n) = 10
- Number of Heads (H) = 7

45/50

90%

Formula:

$$p = \frac{\text{Number of Heads}}{\text{Total Tosses}}$$

Calculation;

$$\hat{p} = \frac{7}{10} = 0.7.$$

The Maximum Likelihood Estimate of the probability of heads is : $p = 0.7$.

Q2. Out of 50 students, 42 passed an Exam. Assuring a Bernoulli distribution, find the probability that a Student passes.

Given:

- Total students = 50
- student passed = 42

Formula:-

$$\hat{p} = \frac{\text{Number of success}}{\text{Total observations}}$$

$$\hat{p} = \frac{42}{50} = 0.84$$

$$\boxed{\hat{p} = 0.84}$$

The estimated probability that a student passes is 0.84.

Question 3:-

A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective. Find the MLE.

Given

Total bulbs = 100

Defective bulbs = 8

Formula;

$$\hat{p} = \frac{\text{Defective bulbs}}{\text{Total bulbs}}.$$

$$\hat{P} = \frac{8}{100} = 0.08$$

$P = 0.08$ ∴ The estimated probability of a defective bulbs is 0.08 (8%).

ustion - 4

A biased die is rolled 60 times. The number 6 appear 18 times. Estimated the probability of rooling a 6 using MLE.

Solution:-

Given:-

- Total rolls = 60
- Number of times 6 appears = 18

Formula:-

$$\hat{p} = \frac{\text{Number of Sixes}}{\text{Total rolls}}$$

$$\hat{p} = \frac{18}{60} = 0.30$$

$p = 0.30$.

The estimed probability of rolling a 6 is 0.30.

Question 5 :-

Given Input (x) = 4

Weight (W) = 2

Actual Output = 10

prediction:

$Y = W \times X$

1. Find the predicted output
2. Calculate the error
3. If the gradient 4 and learning 0.1 find the update weight

Formula:-

$$y = w \times x$$

- Gradient = 4
- Learning Rate = 0.01

i) Predicted output:-

$$y = 2 \times 4 = 8$$

predicted output = 8

ii) calculate the Error:-

formula:-

Error = Actual Output - predicted output

$$10 - 8 = 2$$

$$\boxed{\text{Error} = 2}$$

iii) Find the updated weight.

Gradient Descent formula:-

$$w_{new} = w - (\text{Learning Rate} \times \text{Gradient})$$

$$w_{new} = 2 - (0.01 \times 4)$$

$$= 2 - 0.04$$

$$= 1.96.$$

$$\boxed{\text{Updated weight} = 1.96}$$